When Accurate Loop Closures Hurt Local Accuracy: Diagnosing and Fixing Factor Weighting in KISS-SLAM on KITTI

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Abstract

Loop closure is expected to improve a SLAM trajectory. Evaluating the open-source LiDAR SLAM system KISS-SLAM on the KITTI odometry benchmark, we find that it reduces global error (ATE) by up to 74% but increases local error, measured by the KITTI translation metric, by up to 51% relative to its own odometry. An audit shows that all accepted closures on the tuning sequences are more accurate than the odometry they replace, so the loss does not come from wrong closures. Disabling closures removes 86–96% of the loss, and the corrections the optimizer applies are uncorrelated with where the odometry actually drifted. The code assigns an identity information matrix to every pose-graph factor, which weights one radian of rotation like one metre of translation and makes heading changes nearly free. Weighting the rotational block 10^3 times more heavily, a two-line change, brings the local error of the most affected tuning sequence from 0.517% to 0.333% (odometry: 0.342%) while reducing its ATE from 0.66 m to 0.36 m. On the held-out test sequence 08, where default loop closure made the trajectory worse than odometry, the change reduces local error from 0.954% to 0.841% and ATE from 2.52 m to 2.13 m (odometry: 2.27 m). All hypotheses and acceptance rules were written down before each run; failed predictions are reported. A replication on the NCLT dataset, with the weight fixed on KITTI and three runs per configuration, does not confirm the fix as a reliable improvement: local error falls on all three test sessions (mean −4.0%), but ATE rises by 14.9% on one of them, so the pre-registered criterion fails. A secondary finding characterizes a structured range artifact in the Gazebo `gpu_lidar` sensor that exceeds real LiDAR noise beyond about 8 m.

1 Introduction

Pose-graph SLAM corrects accumulated odometry drift by adding loop-closure constraints and re-optimizing the graph. Global consistency usually improves, but the way the correction is distributed along the trajectory depends on how the factors are weighted. Learning-based systems such as MAC-VO [6] show that metrics-aware covariances in the pose graph improve visual odometry; many LiDAR pipelines, however, still use fixed or identity weights.

This report studies KISS-SLAM [3], which combines KISS-ICP odometry [2] with density-map loop closures and pose-graph optimization. The contributions are:

1. a measurement showing that default loop closure trades local accuracy for global accuracy on several KITTI sequences, even when every closure is accurate;
2. a step-by-step diagnosis that rules out closure errors, the odometry front-end, and the optimizer's iteration limit, and points to identity factor weighting;
3. a two-line fix (rotation-weighted information matrices) evaluated with a tuning/validation/test split, including a held-out sequence on which it turns a net loss into a net gain;
4. a characterization of a range artifact in the Gazebo GPU LiDAR simulator, relevant to anyone using it for noise modelling.

2 Setup and methodology

Data and software. KITTI odometry [1] sequences 00–10 (Velodyne HDL-64E, ground truth available). KISS-ICP 1.3.0 and KISS-SLAM 0.0.2 installed via `pipx`; evaluation with `evo` 1.37.1 [5]; Ubuntu 24.04, RTX 4060 laptop GPU, 32 GB RAM. KISS-SLAM's pose graph is solved with a g2o-based optimizer [4].

Metrics. ATE: RMSE of absolute position error after SE(3) alignment (`evo_ape -a`). RPE: RMSE of translational error over 100 m segments (`evo_rpe -d 100`). KITTI translation error: the benchmark's average over 100–800 m segments, in percent. ATE measures global consistency; RPE and the KITTI metric measure local accuracy.

Split and discipline. Sequences 00–05 were used for tuning, 06–07 for validation, and 08–10 for a single final test. Before every experiment, the hypothesis, the predicted outcome and the acceptance rule were committed to a decision log; results were then recorded whether or not they confirmed the prediction. KISS-ICP was verified to be deterministic (a parallel rerun reproduced all metrics exactly). KISS-SLAM is deterministic on 05, 06, 07 and 08–10, but on 02 the number of accepted closures flips between 1 and 2 across runs (ATE spread 3.3%), and 00 shows a 1.1% ATE spread; differences below these bands are treated as noise.

3 Baselines

A sweep of KISS-ICP preprocessing parameters (deskewing, minimum and maximum range, voxel size) found no configuration that improved mean RPE on the tuning sequences by the pre-set 2% margin; the defaults are close to optimal on KITTI. Table 1 compares KISS-ICP odometry with default KISS-SLAM.

	ATE (m)		KITTI trans. error (%)			
Seq.	Odometry	SLAM	Odometry	SLAM	Change	Closures
00	3.501	0.907	0.528	0.573	+8.5%	7
01	3.149	3.117	0.786	0.798	+1.5%	0
02	6.463	1.854	0.537	0.580	+8.0%	2
03	0.515	0.528	0.677	0.674	−0.4%	0
04	0.241	0.243	0.385	0.384	−0.3%	0
05	1.491	0.661	0.342	0.517	+51%	5
06	0.418	0.328	0.281	0.325	+16%	4
07	0.315	0.315	0.375	0.386	+2.9%	1
08	2.266	2.524	0.820	0.954	+16%	5
09	1.308	1.259	0.534	0.542	+1.5%	0
10	0.709	0.718	0.512	0.525	+2.5%	0

Table 1: KISS-ICP odometry versus default KISS-SLAM. Loop closure reduces ATE by 56–74% on 00, 02 and 05, but increases local error on every sequence with several closures. On 08 it also worsens ATE.

4 Diagnosis

Each step below tested one explanation for the local-accuracy loss, with the prediction fixed in advance.

Are the closures wrong? No. For each of the 14 closures accepted on 00, 02 and 05, the relative pose was compared with ground truth and with the relative pose implied by odometry between the same scans. Every closure was more accurate than the odometry chain it replaces (closure errors 0.12–4.88 m against 0.91–18.1 m), and the overlap score used for acceptance did not visibly predict closure error.

Is it the odometry inside KISS-SLAM? No. With closures disabled (overlap threshold 1.0), KISS-SLAM matches standalone odometry: KITTI error 0.349% on 05 (odometry 0.342%, default 0.517%), 0.532% on 00 and 0.543% on 02. Between 86% and 96% of the gap appears only when closures are applied.

Is the optimizer stopping early? No. Raising the iteration limit from 10 to 50 or 100 left 05 unchanged to six digits.

Where do the corrections go? For each 100 m segment we compared the change introduced by the optimizer with the segment's odometry error. The correlation between correction size and odometry error was 0.13 (05), −0.11 (00) and 0.23 (02); on 05 the root-mean-square correction per segment (0.49 m) was about seven times larger than the evenly distributed drift (about 0.07 m). Removing three of the five closures on 05 shrank the corrections in the affected span by 91% and restored odometry-level local accuracy, but the same intervention had no effect on 00 and 02, so closure selection alone is not a general explanation.

Mechanism. In the KISS-SLAM source, every factor (between local maps, loop closures, and the final per-scan refinement) is added with `np.eye(6)`. Odometry over 100 m drifts on the order of 0.5 m in translation but only about 0.1° (0.0017 rad) in heading, so a physically consistent weighting would make rotation roughly $(0.5/0.0017)^2 \approx 10^5$ times more expensive than an identity matrix does. With identity weights, the optimizer can satisfy a closure by bending headings almost for free, distributing the correction where the odometry was in fact accurate.

5 Fix and evaluation

The patch reads the diagonal of each information matrix from an environment variable and falls back to `np.eye(6)` when unset (verified to reproduce the default exactly on 05). The rotational block is weighted by W: $\Lambda = \mathrm{diag}(1, 1, 1, W, W, W)$, applied to all factors.

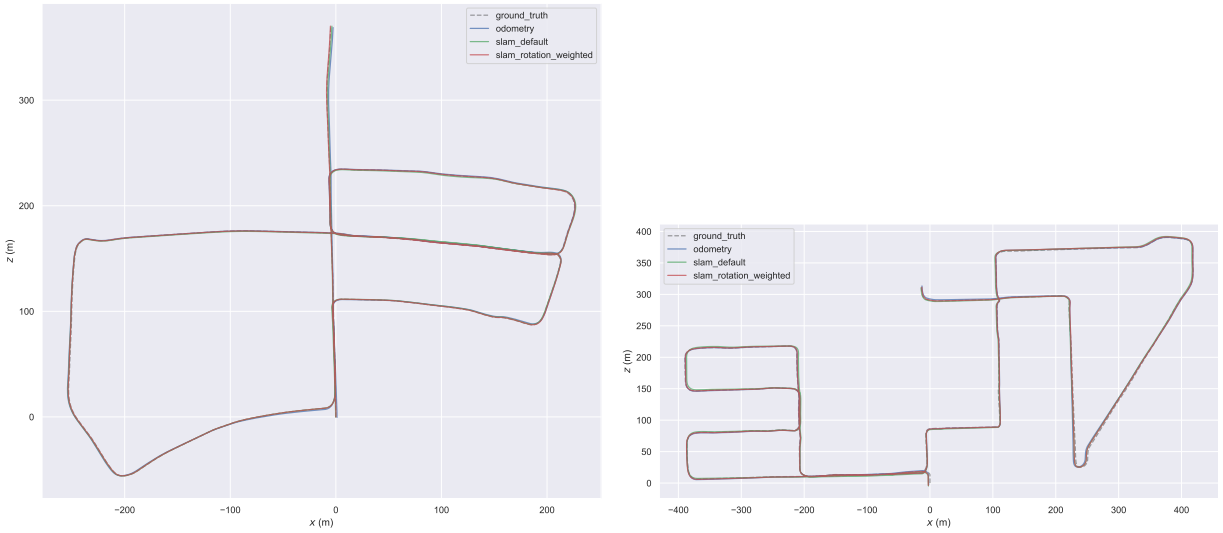
Round 1 (00, 02, 05). Weighting the last three entries by 10^4 reduced 05's KITTI error to 0.341%; weighting the first three by 10^4 instead increased it to 0.725%, which identifies the last block as rotation. The prediction that only one of the two variants would matter was wrong: only the ratio between the blocks matters, so the second variant made rotation cheaper still. Uniformly down-weighting closures by 10 also helped 05 (0.392%) but raised 02's ATE by 17%, outside its noise band.

Weight selection (00–05). The selection rule, fixed in advance, picked the $W \in \{10^3, 10^4, 10^5\}$ with the largest mean improvement in KITTI error, subject to no sequence losing more than 20% ATE or 5% KITTI error. Mean changes were −5.5%, −4.7% and −4.7%; $W = 10^3$ was selected. Loop-free sequences 01, 03 and 04 were unchanged. On 05, $W = 10^3$ gives 0.333% (odometry 0.342%) and ATE 0.364 m (default 0.661 m); the cost on 00 is +1.9%. The gain was already saturated at the smallest weight tested, so the optimum may lie lower; this was not explored to avoid post-hoc tuning.

Validation and test. Table 2 reports the single runs on validation and test sequences. Both validation sequences pass the pre-set rule. On the held-out sequence 08, default loop closure made the trajectory worse than odometry in both metrics; with $W = 10^3$ it becomes better than odometry in ATE and loses only 2.6% in local accuracy. Sequences 09 and 10 have no accepted closures and are unchanged. Repeated default runs on 08–10 were identical, so the 08 difference is not run-to-run noise.

		KITTI trans. error (%)			ATE (m)		
Role	Seq.	Odometry	Default	$W = 10^3$	Odometry	Default	$W = 10^3$
Validation	06	0.281	0.325	0.332	0.418	0.328	0.331
Validation	07	0.375	0.386	0.391	0.314	0.315	0.314
Test	08	0.820	0.954	**0.841**	2.266	2.524	**2.126**
Test	09	0.534	0.542	0.542	1.309	1.259	1.256
Test	10	0.512	0.525	0.525	0.710	0.718	0.718

Table 2: Validation and test results, each run once after W was fixed on the tuning sequences.



(a) Sequence 05 (tuning)

(b) Sequence 08 (test)

Figure 1: Top-down trajectories (x–z plane): ground truth, KISS-ICP odometry, default KISS-SLAM and rotation-weighted KISS-SLAM ($W = 10^3$), aligned to ground truth.

6 Secondary finding: a range artifact in Gazebo's GPU LiDAR

To model sensor noise for simulation, range noise was measured on KITTI by second differences along each beam on road surface, an estimator verified to within 1% on synthetic white noise. It gives 0.84–0.90 cm at 3–10 m and 1.65–1.83 cm at 30–40 m (sequences 00, 02, 05), an upper bound on white range noise. Plane-fit residuals, by contrast, are 4–8 cm and measure road shape rather than the sensor.

In Gazebo Sim 8 (Harmonic), a 64-beam `gpu_lidar` over a flat ground plane showed two artifacts. A 360° configuration returned spurious points 1–1.6 m from the sensor over most azimuths; a 120° sector did not. Within the clean sector, the ideal (noise-free) ranges contain deterministic jumps: about 40% of ground triples include a step, 88% of steps appear as adjacent opposite-sign pairs, and the step size is about 0.19% of range beyond 8 m. Changing the sensor's minimum range had no effect. The resulting second-difference spread (2.6 cm at 9 m, 15 cm at 29 m) exceeds the real sensor noise beyond about 8 m, by roughly ten times at 30 m, so adding a realistic noise model

on top of this sensor does not produce realistic data at range. The cause of the jumps was not identified. A noise-injection ROS 2 node was implemented and verified (injected and measured noise agree within 2%).

7 Replication on NCLT

Why and how. KITTI offers few loops and one platform. The same comparison was repeated on the University of Michigan North Campus Long-Term dataset (NCLT) [8]: a Segway with a Velodyne HDL-32E driving the same campus in sessions over 15 months. The split was fixed before downloading: tuning 2012-01-15, 2012-04-29, 2012-08-04; validation 2012-05-26, 2012-10-28; test 2012-02-12, 2012-06-15, 2013-04-05, run once. $W = 10^3$ was taken from KITTI and *not* re-tuned. NCLT ground truth comes from a LiDAR and RTK-GPS graph, so it partly shares the LiDAR being evaluated. Metrics and software are as in Section 2 (a NaN-tolerant re-implementation of the metrics reproduces `evo` for ATE and the KITTI metric).

Noise. Unlike on KITTI, KISS-SLAM is far from repeatable on NCLT: with the same number of accepted closures, ATE ranged from 2.0 to 6.4 m on 2012-04-29. KISS-ICP itself was bit-identical across runs on the session where it was checked, so the spread comes from the SLAM side. Every configuration was therefore run three times, and a difference counts only if the three-run ranges do not overlap. A six-run recheck on 2012-05-26 after an operating-system upgrade showed that three runs had underestimated the spread, and one validation claim was retracted.

Tuning and validation. Default loop closure lowered ATE on 8 of 9 tuning runs, but raised local error clearly on only one session (2012-04-29, all three runs), against two of three predicted; the KITTI effect does not transfer cleanly. On 2012-04-29, $W = 10^3$ lowered both metrics beyond noise, as predicted (Table 3). Both validation sessions passed the pre-set rule (not worse beyond noise).

		KITTI trans. error (%)		ATE (m)	
Role	Session	Default	$W = 10^3$	Default	$W = 10^3$
Tuning	2012-01-15	1.395–1.397	1.373–1.413	2.32–3.08	2.71–3.08
Tuning	2012-04-29	1.415–1.474	**1.398–1.408**	2.01–6.37	**1.74–2.00**
Tuning	2012-08-04	1.638–1.887	1.666–1.762	2.92–5.96	2.60–5.42
Validation	2012-05-26*	1.333–1.422	1.293–1.383	2.39–3.76	1.44–2.59
Validation	2012-10-28	1.369–1.394	1.344–1.371	1.71–3.12	1.70–2.04
Test	2012-02-12	1.609–1.715	**1.563–1.601**	2.87–3.61	3.04–3.80
Test	2012-06-15	1.239–1.266	1.196–1.253	2.03–2.05	*2.06–2.57*
Test	2013-04-05	2.089–2.130	1.775–2.431	9.19–9.78	4.92–14.37

Table 3: KISS-SLAM on NCLT, default versus $W = 10^3$: ranges over three runs (*six runs). Bold: better beyond noise; italic: worse beyond noise. KISS-ICP odometry on the tuning sessions: KITTI 1.48, 1.38, 1.74%; ATE 9.85, 4.61, 30.9 m.

Test. The registered criterion required $W = 10^3$ to lower the mean KITTI metric over the test sessions without raising any session's mean ATE by more than 10%. The first part holds (1.674% to 1.606%, lower on all three sessions); the second fails on 2012-06-15, where mean ATE rose by 14.9% (2.03 to 2.34 m), a small absolute change on a session where the default happened to be very repeatable. On 2013-04-05 the weighted runs were far more spread than the default ones (ATE 4.9–14.4 m). The criterion is therefore not met: rotation weighting chosen on KITTI does not transfer to NCLT as a reliable improvement, although it does not clearly hurt local accuracy and may help it slightly.

A unit convention. Rebuilding KISS-SLAM's pose graph in GTSAM [9] for the next phase showed that g2o's SE(3) edge measures rotation error by the quaternion vector, about half the angle. The weight $W = 10^3$ is therefore 250 on the rotation angle; the physical estimate of Section 4 ($\approx 10^5$ on the angle) corresponds to $\approx 4 \times 10^5$ in g2o's convention. With this conversion, GTSAM reproduces all 18 KISS-SLAM tuning graphs to within 1 cm.

8 Limitations

The KITTI results come from one sensor and one SLAM system; on NCLT the fix did not pass its test criterion, so the KITTI improvement should not be read as general. The KITTI effect is large on two sequences (05 and 08) and small or absent elsewhere; the local loss on 06 is not recovered. W was chosen from three candidates and the information matrix is a scaled identity, not a calibrated covariance. KISS-SLAM is not fully deterministic on some sequences. The simulator findings concern one Gazebo version and a flat-ground statistic.

9 Conclusion and future work

Identity information matrices let a pose-graph optimizer absorb loop-closure corrections as cheap heading changes, which costs local accuracy even when every closure is accurate. Weighting rotation consistently with the physical scale of odometry error removes most of that cost on the affected KITTI sequences, including a held-out one, while improving global consistency. On NCLT, a single fixed weight chosen on KITTI lowers local error slightly but does not reliably keep global accuracy, which points to weights that depend on the factor rather than one constant. Natural next steps are per-factor covariances estimated from the registration itself, evaluation on datasets with more loops and more degenerate scenes [7], and learned covariances where the classical estimates fail.

Reproducibility. Code, configurations, the decision log with every prediction and result, and the patch are available at `https://github.com/yassinmtibaa/lidar-loop-closure-weighting`.

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